

SYDNEY TECHNICAL HIGH SCHOOL



EXTENSION 1 MATHEMATICS PRELIMINARY ASSESSMENT TASK 2 2010

Time Allowed: 70 minutes

Instructions:

- Write using blue or black pen.
- Approved calculators may be used.
- Attempt all questions.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- PLEASE START EACH NEW QUESTION ON A NEW PAGE

Name: _____ **Teacher:** _____

Q1	Q2	Q3	Q4	Q5	Q6
/10	/10	/10	/10	/10	/10

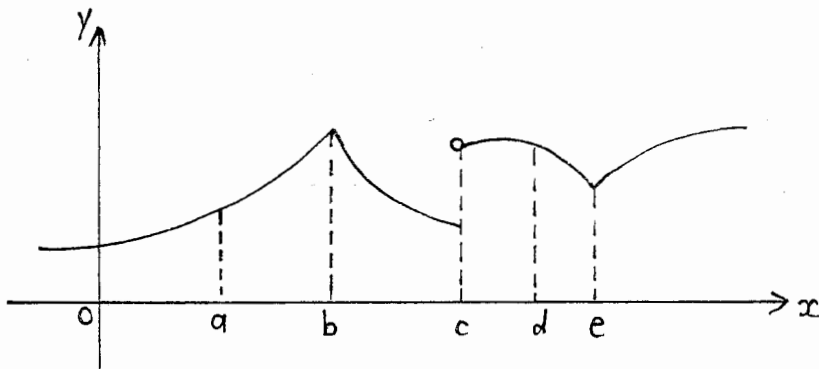
Total
/60

Question 1

- a) Find the values of k such that $x^2 + (k + 1)x + 4 = 0$ has real roots. 3
- b) Evaluate $\lim_{x \rightarrow 3} \frac{2x^2 + x - 21}{x^2 - 3x}$ 2
- c) (i) Differentiate $y = \sqrt{2x - 1}$ 2
- (ii) Hence find the equation of the normal to the above function at $x = 1$. 3

Question 2

- a) If $\sin x = \frac{2}{3}$ and $\cos y = \frac{1}{9}$, where angles x and y are acute, show
- (i) $\cos(x - y) = \frac{\sqrt{5}}{3}$ 2
- (ii) $\cos y - \cos 2x = 0$ 2
- b) Find the acute angle between the lines $x = 2$ and $y = \frac{1}{\sqrt{3}}x + 2$ 2
- c) Let $t = \tan \frac{\theta}{2}$ and then use the 't' results to prove 3
- $$\sec \theta - \tan \theta = \frac{1}{\sec \theta + \tan \theta}$$
- d) 1



For which x value(s) is the function not differentiable?

Question 3**Marks**

- a) If α and β are the roots of the equation $2x^2 - 13x + 9 = 0$,
- (i) Show that $(\alpha + 1)(\beta + 1) = 12$ 2
- (ii) Find the quadratic equation with integer coefficients which has roots $\alpha + 1$ and $\beta + 1$. 2
- b) Show that the line $y = mx - m^2$ is a tangent to the parabola $y = \frac{x^2}{4}$ for all real m . 2
- c) Use differentiation by first principles to find the derivative of $f(x) = x^2 - 5x$. 3
- d) If the discriminant of a quadratic equation has the value of 36, what conclusion can you make about the 2 distinct real roots of the equation? 1

Question 4

- a) (i) Expand $\tan(A + B)$ 1
- (ii) Find the exact value of $\tan 105^\circ$ (do not rationalise denominator) 2
- b) Differentiate $y = \frac{1+2x}{1-2x}$ 2
- and give your answer as a fully simplified algebraic fraction.
- c) If the difference between the roots $2x^2 - 12x + p + 2 = 0$ is equal to 2, find the value of p . 3
- d) At what point(s) on the curve $y = x^2 - 2x - 1$ is the tangent parallel to the line $y = x$. 2

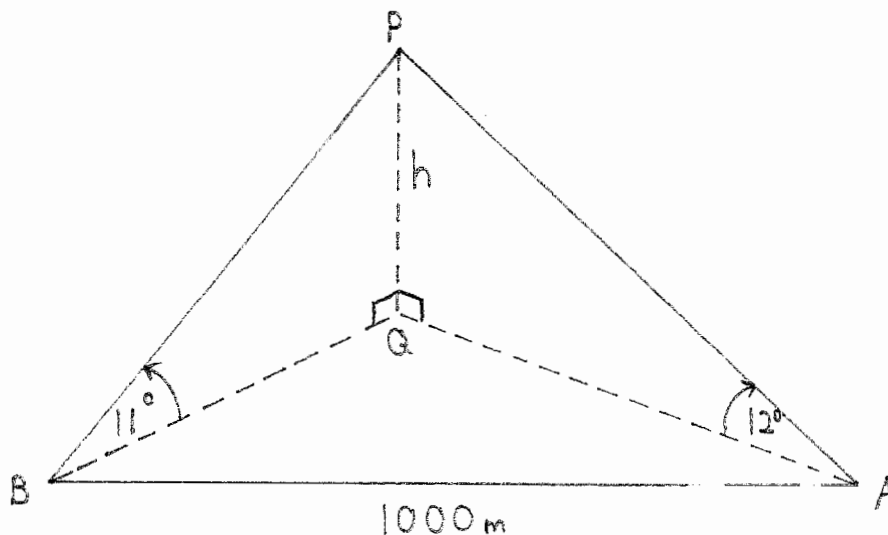
Question 5**Marks**

- a) Solve $2\cos^2 x = \sin 2x$ for $0 \leq x \leq \pi$. 3
- b) Find the ratio in which the point $(\frac{7}{9}, \frac{-2}{9})$ divides the interval joining $(3, -6)$ and $(-2, 7)$. 3
- c) (i) Express $\cos x + \sqrt{3}\sin x$ in the form $A\cos(x - \alpha)$ where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$. 2
- (ii) Hence solve $\cos x + \sqrt{3}\sin x = 1$ for $-\pi \leq x \leq \pi$. 2

Question 6

Marks

- a) Solve $9^x - 7(3^x) - 18 = 0$ 2
- b) Show that $(p + 3)x^2 - px + 1$ can never be negative definite. 3
- c)



The angle of elevation of a tower PQ , of height h metres, at a point A due east of it is 12° . From another point B , the bearing of the tower is $015^\circ T$ and the angle of elevation is 11° . The points A and B are 1000m apart and on the same level as the base Q of the tower.

- (i) By sketching a top view diagram show that $\angle AQB$ is 105° . 1
- (ii) Show that $AQ = h \tan 78^\circ$. 1
- (iii) Calculate h to the nearest metre. 3

Prelim. Ass. Task 2 - Extension 1 - 2010Question 1a) Real roots if $\Delta \geq 0$

$$(k+1)^2 - 4 \times 1 \times 4 \geq 0$$

$$k^2 + 2k + 1 - 16 \geq 0$$

$$k^2 + 2k - 15 \geq 0$$

$$(k-3)(k+5) \geq 0$$

$$k \geq 3, k \leq -5$$

b) $\lim_{x \rightarrow 3} \frac{2x^2 + x - 21}{x^2 - 3x}$

$$\lim_{x \rightarrow 3} \frac{(2x+7)(x-3)}{x(x-3)}$$

$$\lim_{x \rightarrow 3} \frac{(2x+7)}{x}$$

$$= \frac{13}{3}$$

c) i) $y = \sqrt{2x-1}$

$$= (2x-1)^{\frac{1}{2}}$$

$$y' = \frac{1}{2} (2x-1)^{-\frac{1}{2}} \times 2$$

c) ii) $y' = \frac{1}{\sqrt{2x-1}}$

At $x=1$

$$y' = \frac{1}{\sqrt{1}} = 1 \text{ (m of tan)}$$

$$\therefore m_{\text{normal}} = -1$$

At $x=1, y=1$

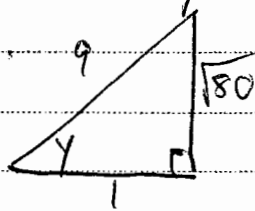
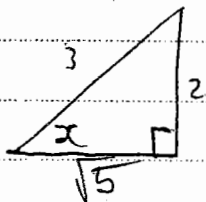
$$\therefore y-1 = -(x-1)$$

$$y = -x + 2$$

or $x + y - 2 = 0$

Question 2

a) ii) $\cos(x-y) = \cos x \cos y + \sin x \sin y$



$$= \frac{\sqrt{5}}{3} \times \frac{1}{9} + \frac{2}{3} \times \frac{\sqrt{80}}{9}$$

$$= \frac{\sqrt{5}}{27} + \frac{2\sqrt{80}}{27}$$

$$= \frac{\sqrt{5} + 2 \times 4\sqrt{5}}{27}$$

$$= \frac{9\sqrt{5}}{27} = \frac{\sqrt{5}}{3}$$

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$$\begin{aligned}
 \text{cii) } \cos y - \cos 2x &= \cos y - (2\cos^2 x - 1) \\
 &= \frac{1}{9} - (2 \times (\frac{\sqrt{5}}{3})^2 - 1) \\
 &= \frac{1}{9} - (\frac{10}{9} - 1) \\
 &= \frac{1}{9} - \frac{1}{9} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } m &= \tan \theta \\
 \frac{1}{\sqrt{3}} &= \tan \theta \\
 \theta &= 30^\circ
 \end{aligned}$$

is angle made with
x axis $\therefore 60^\circ$ is
the angle made
with $x=2$.

$$\begin{aligned}
 \text{c) } \sec \theta - \tan \theta &= \frac{1}{\sec \theta + \tan \theta} \\
 \frac{1+t^2}{1-t^2} - \frac{2t}{1-t^2} &= \frac{1}{\frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2}} \times \frac{1-t^2}{1-t^2} \\
 \frac{1+t^2-2t+1}{1-t^2} &= \frac{1-t^2}{1+t^2+2t} \\
 \frac{(t-1)^2}{(1-t)(1+t)} &= \frac{(1-t)(1+t)}{(1-t)(1+t)} \\
 \frac{(1-t)(1+t)}{(1-t)(1+t)} &= \frac{(1-t)(1+t)}{(1-t)(1+t)} \\
 \frac{1-t}{1+t} &= \frac{1-t}{1+t}
 \end{aligned}$$

$$\text{d) } x = b, c, e$$

Question 3

$$\begin{aligned}
 \text{a) i) } (\alpha+1)(\beta+1) &= \alpha\beta + (\alpha+\beta) + 1 \\
 &= \frac{9}{2} + \frac{-13}{2} + 1 \\
 &= 4\frac{1}{2} + 6\frac{1}{2} + 1 \\
 &= 12
 \end{aligned}$$

$$\begin{aligned}
 \text{cii) } x^2 - (\text{sum of roots}) + \text{product} & \\
 \text{Sum of roots is} & \\
 \alpha + 1 + \beta + 1 &= 2 + \frac{13}{2} = 8\frac{1}{2} \\
 \therefore x^2 - 8\frac{1}{2}x + 12 &= 0 \\
 2x^2 - 17x + 24 &= 0
 \end{aligned}$$

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b) Solve simultaneously.

Tangent if $\Delta = 0$

$$mx - m^2 = \frac{x^2}{4}$$

$$4mx - 4m^2 = x^2$$

$$x^2 - 4mx + 4m^2 = 0$$

$$\Delta = (-4m)^2 - 4 \times 1 \times 4m^2$$

$$= 16m^2 - 16m^2$$

$$\Delta = 0$$

 $\therefore$ Line is a tangent.

$$c) f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^2 - 5(x+h) - (-x^2 + 5x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 5x - 5h + x^2 - 5x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 5)}{h}$$

$$= 2x - 5$$

d) They are rational

Question 4

$$a) \text{ c) } \tan(A+B)$$

$$= \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan 105 = \tan(60 + 45)$$

$$= \frac{\tan 60 + \tan 45}{1 - \tan 60 \tan 45}$$

$$= \frac{\sqrt{3} + 1}{1 - \sqrt{3} \times 1}$$

$$= \frac{\sqrt{3} + 1}{1 - \sqrt{3}}$$

$$b) y = \frac{1+2x}{1-2x}$$

$$y' = \frac{(1-2x) \times 2 - (1+2x) \times (-2)}{(1-2x)^2}$$

$$= \frac{2-4x+2+4x}{(1-2x)^2}$$

$$= \frac{4}{(1-2x)^2}$$

$$c) 2x^2 - 12x + p + 2 = 0$$

Let roots be $\alpha, \alpha+2$

$$\alpha + \alpha + 2 = \frac{12}{2} \quad \alpha(\alpha+2) = \frac{p+2}{2}$$

$$2\alpha + 2 = 6$$

$$\alpha = 2$$

$$\therefore 2(2+2) = \frac{p+2}{2}$$

$$16 = \frac{p+2}{2}$$

$$p = 14$$

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$$\begin{aligned}
 d) y &= x^2 - 2x - 1 \\
 y' &= 2x - 2 = 1 \quad (\text{gradient of } y = x) \\
 2x &= 3 \\
 x &= \frac{3}{2} \\
 y &= \left(\frac{3}{2}\right)^2 - 2 \times \frac{3}{2} - 1 \\
 &= \frac{9}{4} - 4 \\
 y &= -1\frac{3}{4} \quad \therefore \text{Point is } \left(\frac{3}{2}, -1\frac{3}{4}\right)
 \end{aligned}$$

Question 5

$$\begin{aligned}
 a) 2\cos^2 x &= \sin 2x \\
 2\cos^2 x &= 2\sin x \cos x \\
 \cos^2 x - \sin x \cos x &= 0 \\
 \cos x (\cos x - \sin x) &= 0 \\
 \cos x = 0 \quad \text{or} \quad \cos x &= \sin x \\
 x = \frac{\pi}{2} \quad \tan x &= 1 \\
 x &= \frac{\pi}{4} \\
 \text{over domain } 0 \leq x &\leq \pi
 \end{aligned}$$

$$\begin{aligned}
 b) \begin{matrix} x_1 & y_1 \\ (3, & -6) \end{matrix} \quad \begin{matrix} x_2 & y_2 \\ (-2, & 7) \end{matrix} \\
 \begin{matrix} x & y \\ (\frac{7}{9}, & \frac{2}{9}) \end{matrix} \quad k:l? \\
 x = \frac{kx_2 + lx_1}{k+l} \\
 \frac{7}{9} = \frac{k(-2) + l(3)}{k+l} \\
 7(k+l) = (-2k+3l)9 \\
 7k+7l = -18k+27l \\
 25k = 20l \\
 \therefore k:l \text{ is } 4:5
 \end{aligned}$$

$$\begin{aligned}
 c) i) \cos x + \sqrt{3} \sin x \\
 A = \sqrt{(\sqrt{3})^2 + 1} = 2 \\
 \alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3} \\
 \therefore = 2 \cos(x - \frac{\pi}{3})
 \end{aligned}$$

$$\begin{aligned}
 c) ii) 2\cos(x - \frac{\pi}{3}) &= 1 \\
 \cos(x - \frac{\pi}{3}) &= \frac{1}{2} \\
 x - \frac{\pi}{3} &= \frac{\pi}{3}, -\frac{\pi}{3} \\
 x &= \frac{2\pi}{3} \quad \text{or} \quad 0
 \end{aligned}$$

Question 6

$$\begin{aligned}
 a) 9^x - 7(3^x) - 18 &= 0 \\
 3^{2x} - 7(3^x) - 18 &= 0 \\
 (3^x)^2 - 7(3^x) - 18 &= 0 \\
 (3^x + 2)(3^x - 9) &= 0 \\
 3^x = -2 \quad \text{or} \quad 3^x &= 9 \\
 x = 2 \quad \text{or} \quad 3^x &= -2 \quad \text{no}
 \end{aligned}$$

$$\begin{aligned}
 d) \quad y &= x^2 - 2x - 1 \\
 y' &= 2x - 2 = 1 \quad (\text{gradient of } y = x) \\
 2x &= 3 \\
 x &= \frac{3}{2} \\
 y &= \left(\frac{3}{2}\right)^2 - 2 \times \frac{3}{2} - 1 \\
 &= \frac{9}{4} - 3 \\
 y &= 5\frac{1}{4} \quad \therefore \text{Point is } \left(\frac{3}{2}, 5\frac{1}{4}\right)
 \end{aligned}$$

Question 5

$$\begin{aligned}
 a) \quad 2\cos^2 x &= \sin 2x \\
 2\cos^2 x &= 2\sin x \cos x \\
 \cos^2 x - \sin x \cos x &= 0 \\
 \cos x (\cos x - \sin x) &= 0 \\
 \cos x = 0 \quad \text{or} \quad \cos x &= \sin x \\
 x = \frac{\pi}{2} \quad \tan x &= 1 \\
 x &= \frac{\pi}{4} \\
 \text{over domain } 0 \leq x &\leq \pi
 \end{aligned}$$

$$\begin{aligned}
 b) \quad (x_1, y_1) &= (3, -6) \quad (x_2, y_2) = (-2, 7) \\
 (x, y) &= \left(\frac{x_1}{9}, \frac{y_1}{9}\right) \quad k:l? \\
 x &= \frac{kx_2 + lx_1}{k+l} \\
 \frac{7}{9} &= \frac{k(-2) + l(3)}{k+l} \\
 7(k+l) &= (-2k + 3l)9 \\
 7k + 7l &= -18k + 27l \\
 25k &= 20l \\
 \therefore k:l &\text{ is } 4:5
 \end{aligned}$$

$$\begin{aligned}
 c) \quad \cos x + \sqrt{3} \sin x \\
 A &= \sqrt{(\sqrt{3})^2 + 1} = 2 \\
 \alpha &= \tan^{-1} \sqrt{3} = \frac{\pi}{3} \\
 \therefore &= 2 \cos\left(x - \frac{\pi}{3}\right)
 \end{aligned}$$

$$\begin{aligned}
 c) \quad 2\cos\left(x - \frac{\pi}{3}\right) &= 1 \\
 \cos\left(x - \frac{\pi}{3}\right) &= \frac{1}{2} \\
 x - \frac{\pi}{3} &= \frac{\pi}{3}, -\frac{\pi}{3} \\
 x &= \frac{2\pi}{3} \text{ or } 0.
 \end{aligned}$$

Question 6

$$\begin{aligned}
 a) \quad 9^x - 7(3^x) - 18 &= 0 \\
 3^{2x} - 7(3^x) - 18 &= 0 \\
 (3^x)^2 - 7(3^x) - 18 &= 0 \\
 (3^x + 2)(3^x - 9) &= 0 \\
 3^x &= -2 \quad \text{or} \quad 3^x = 9 \\
 x &= 2 \quad \text{or} \quad 3^x = -2 \text{ no solution}
 \end{aligned}$$

b) $(p+3)x^2 - px + 1$
negative definite
if

$$p+3 < 0$$

$$\text{and } \Delta < 0$$

$$p < -3$$

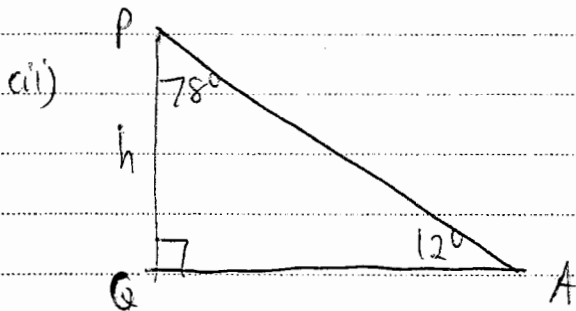
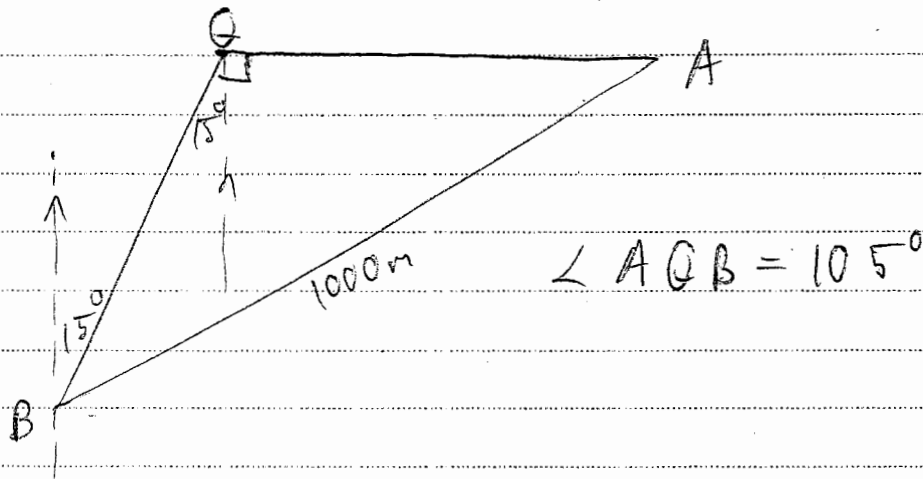
$$p^2 - 4(p+3) < 0$$

$$p^2 - 4p - 12 < 0$$

$$(p+2)(p-6) < 0$$

$\therefore$ Not possible and so can't be negative definite.

c)



$$\tan 78 = \frac{AQ}{h}$$

$$AQ = h \tan 78$$

cii) Similarly

$$BQ = h \tan 79$$

$$\text{By the cosine rule}$$

$$AB^2 = AQ^2 + BQ^2 - 2 \times AQ \times BQ \times \cos 105^\circ$$

$$1000^2 = h^2 \tan^2 78 + h^2 \tan^2 79 - 2h^2 \tan 79 \tan 78 \cos 105^\circ$$

$$h^2 = \frac{1000000}{\tan^2 78 + \tan^2 79 - 2 \tan 78 \tan 79 \cos 105^\circ}$$

$$h = 128 \text{ m}$$